

SHRISH RAHUL HUKKERI

Computer Science Engineer · AI / ML · LLMs · IoT & Embedded Systems · AIoT · IoV · Edge AI

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PROFESSIONAL SUMMARY

Final-year Computer Science Engineer @ RNS Institute of Technology with hands-on experience across AI/ML, Gen AI, LLMs, IoT, Robotics, Electronics & software-hardware stacks. Professional & strong communication skills, class representative, teambuilder, mentoring multiple R&D projects, experienced in teaching & coaching with valuable internship exposure from Samsung Innovation Campus and Pantech. Championed multiple national-level hackathons and innovation awards.

EDUCATION

Bachelor of Engineering — Computer Science & Engineering · *RNS Institute of Technology, Bengaluru*
CGPA: 9.07 / 10 (6th semester) · Relevant coursework: DSA, DAA, DBMS, Computer Networks, OS, AI/ML, Software Engineering

EXPERIENCE

Research Assistant — CSE Dept. · *RNSIT Bengaluru, India | Aug 2025 – Apr 2026*

- Selected as the first-ever Research Assistant from the CSE Department at RNSIT, working directly under research faculty on computer vision & affective computing.

IoT Intern · *Samsung Innovation Campus @ RNSIT Center of Excellence Bengaluru, India | Sep 2025 – May 2026*

- Ranked Top 15 out of 450+ engineers in a nationally competitive Samsung-backed IoT program covering cloud-connected IoT, sensor integration, embedded systems and end-to-end solution architecture.
- Designed and implemented IoT prototypes integrating cloud dashboards, real-time sensor data pipelines and device-to-cloud communication using industry-standard protocols.

AI & Robotics Intern · *Pantech.AI Bengaluru, India (Remote) | Jun 2026*

- Implemented computer vision pipelines for real-time YOLO-based object detection, lane detection and traffic sign recognition achieving >90% detection accuracy on benchmark datasets.
- Explored LiDAR sensor fusion, SLAM algorithms and Raspberry Pi-based autonomous robot development integrating ML inference at the edge.

Embedded Systems Intern · *Pantech.AI Bengaluru, India (Remote) | Jun 2025*

- Developed 5+ Arduino and ESP32-based embedded prototypes integrating multi-sensor inputs, actuator control and hardware-software co-design for rapid IoT application prototyping.

Head — Drone & UAV Development Team · *Zepco & CDDT, RNSIT | 2025 – Present*

- Contributing to UAV design, aeromodelling & Drones across embedded control systems, sensor fusion, ESC calibration and hardware-software integration. Among Top 14 at Boeing Zonal Aeromodelling Competition. 1st Place at SVIT Drone Expo.

Technical Head · *Antariksh Space Club, RNSIT | 2026 – Present*

- Contributing to technical supervision and expertise in Web development, Embedded Systems, IoT, Robotics, GIS, Satellite Data Processing & Computer Vision.

PROJECTS

Smart Automation for Child Safety - SAV4Secure
Stack: AirTag · SPLASH Programming · Smart Automations · Shortcuts

- Architected an AirTag-driven child safety monitoring automation winning 1st Place at Smart India Hackathon, the nation's largest student hackathon platform.

FAST — Farmers Assistance System using Technology with AI in Food Chain
Stack: Arduino Uno · Soil/Temperature Sensors · Python · IoT Dashboard · ML Classification

- Developed a precision-agriculture decision support system integrating soil moisture, temperature and humidity sensor feeds with an ML classifier recommending crop actions (irrigation, fertilisation, harvest timing).

Secure Assessment & Question Paper Generation Platform
Stack: Python · Flask · PostgreSQL · SQLAlchemy · HTML · CSS · JavaScript · Role-Based Access Control (RBAC) · PDF Generation

• Designed a secure desktop examination management system that centralizes question-bank submission, examination paper generation, approval workflows, and controlled paper distribution, with the Controller of Examinations (CoE) retaining complete administrative authority, inspired by the need of the hour accountability and robust systems after NEET paper leaks.

React Native Mobile Application Development

Stack: React Native · TypeScript · Expo · Supabase · REST APIs · State Management

• Crafted high-performance, cross-platform iOS and Android apps using React Native. Designed responsive UIs, offline-first capabilities, and secure REST API integrations for scalable mobile experiences.

AI-Enabled Unified DevOps Intelligence Platform (Sentinel)

Stack: Python · Flask · React · SQLite · Local LLM (LM Studio) · Edge Voice AI

• Built a unified SDLC and DevOps control plane that keeps requirements, QA, releases, service health and meetings in one data model. A local language model runs on the organization's own machine and can only name a tool from a role-filtered set of 13 approval-gated tools. API Hub with 23 connectors (GitHub, GitLab, Jira, Jenkins, Kubernetes, Datadog, etc.)

TECHNICAL SKILLS

Programming Languages: Python (primary), C, C++, TypeScript, JavaScript, JSX/TSX, SQL, HTML5, CSS3, Shell Scripting, LaTeX

Frontend & Web Development: React, Tailwind CSS, Monaco Editor, Recharts, Chart.js, responsive UI development, component-based architecture, PWAs, Electron, REST API integration, WebSockets

3D Graphics: Three.js, React Three Fiber, browser-based 3D rendering, GLB/glTF asset workflows

Mobile Development: React Native, Expo, React Navigation, NativeWind, Reanimated, mobile API integration, Android/iOS build workflows, APK and IPA generation, AR & VR Technologies

Backend & APIs: Python, Flask, FastAPI, Node.js, Express.js, REST APIs, WebSockets, API integration, authentication and authorization, background jobs, real-time application workflows

AI & Machine Learning: PyTorch, Torchvision, OpenCV, MediaPipe, MediaPipe Face Mesh, XGBoost, Hugging Face, Kaggle, feature engineering, model evaluation, object detection, facial expression recognition, posture and gesture analysis

LLMs, VLMs, Speech & Generative AI: Ollama, LM Studio, Groq integration, ONNX Runtime, Whisper-based speech recognition, text-to-speech integration, retrieval-augmented generation (RAG), embeddings, prompt engineering, local AI inference.

Embedded Systems & Hardware: Arduino, ESP32, C/C++ firmware, sensor integration, serial communication, hardware-software integration, IoT prototypes, NFC/RFID projects, embedded-system debugging, end-to-end IoT System Design Architecture & Cloud

DevOps & Software Quality: Git, GitHub, Docker, Docker Compose, Linux, CI/CD fundamentals, GitHub Actions workflows, automated testing, pytest, ESLint, static analysis, containerized applications, application health checks

Security & Networking: TCP/IP fundamentals, IPv4/IPv6, TCP/UDP, routing fundamentals, access control, secure authentication, cryptography fundamentals, secret management, DDoS detection and mitigation concepts, Moving Target Defense (MTD) research

Development & Research Tools: VS Code, Arduino IDE, Jupyter Notebook, Google Colab, Antigravity, Vercel, Render, Canva, Penpot

Wireless & Embedded Protocols: LoRa, BLE, ESP-NOW, GPS/GNSS, MQTT, HTTP-based IoT communication, ArduPilot, MAVLink, USB HID and XInput/gamepad firmware

Currently Learning: Swift, Swift UI, Databases & Caching, DevOps & Infrastructure**,** Networking & Cybersecurity Tools

AWARDS & ACHIEVEMENTS

• Smart India Hackathon Winner — ₹25,000 prize for Airtag-driven child safety automation.
• Top 15 / 450+ engineers — Samsung Innovation Campus IoT Internship.
• Top 14 / Zonal Level — Boeing Aeromodelling Competition (UAV design and flight performance).
• 1st Place — SVIT Drone Expo, Project Open House Panorama 2024 & 2025, Kaggle Deep Learning Competition
• 2nd Place — Luminus Solaris Grand Hackathon, Science Idea Tank; 3rd Place — Vigyanotsava.
• Finalist — PESU Ignite IoT Hackathon; Best Student-Teacher Award (Shikshak Parv).
• Designated as top certification performer with 140+ certifications across Infosys Springboard, NPTEL, NSDC, IBM, Cisco & more

PUBLICATIONS

ICECST International IEEE Conference | Lincoln University College, Petaling Jaya, Malaysia | *30th September 2026* : On-Node RSSI Ranging to Reduce Airtime in Large-Scale LoRaWAN Asset and Vehicle Tracking

INCCSC International IEEE Conference | LBSITW, Kerala, India | *5th October 2026* : **HUMAN**: A Hybrid Unsupervised Multi-Aspect Network for Explainable Pattern Discovery in Heterogeneous Tabular Data

CERTIFICATIONS

140+ certifications across IoT, AI/ML, LLMs, VLMs, computer vision, data science, networking and professional skills